Online Appendices
Formulation Details of Alternative Methods

1 Details of the Column Generation Formulation

Let $\mathcal{C}_r$ denote the set of all feasible columns, namely sequences, for runway $r \in \mathcal{R}$, where each sequence $c \in \mathcal{C}_r$ encodes a complete ordering of the aircraft assigned to runway r together with the associated runway operation times. Let λ_c be a binary variable equal to 1 if sequence c is selected, and let w_c be the delay cost of sequence c. The master problem is a set partitioning problem of the following form.

$$\min \quad \sum_{r\in\mathcal{R}} \sum_{c\in\mathcal{C}_r} w_c \, \lambda_c, \tag{1}$$

$$\text{s.t.} \quad \sum_{c\in\mathcal{C}_r} \lambda_c = 1, \qquad \forall r \in \mathcal{R}, \tag{2}$$

$$\sum_{c\in\mathcal{C}_{r(j)}} u_j^c \, \lambda_c - \sum_{c\in\mathcal{C}_{r(i)}} u_i^c \, \lambda_c \geq \zeta_p + \Pi_{ij} \left(1 - K_{ij}^p\right), \qquad \forall p \in \mathcal{P}, \forall i,j \in \mathcal{D}_p : i \neq j, \tag{3}$$

$$K_{ij}^p + K_{ji}^p = 1, \qquad \forall p \in \mathcal{P}, \forall i,j \in \mathcal{D}_p : i \neq j, \tag{4}$$

$$\lambda_c \in \{0,1\}, \qquad \forall c \in \mathcal{C}_r, \tag{5}$$

$$K_{ij}^p \in \{0,1\}, \qquad \forall p \in \mathcal{P}, \forall i,j \in \mathcal{D}_p : i \neq j, \tag{6}$$

where $r(i)$ denotes the runway to which aircraft i is assigned, and u_i^c is the pushback time of departure i in sequence c, which is a known parameter of the column since the take-off time is encoded in the sequence and the taxi time is given. Constraint (2) ensures that exactly one sequence is selected for each runway. Constraint (3) enforces the ramp-level pushback interval, in which the pushback times $\sum_{c\in\mathcal{C}_{r(i)}} u_i^c \lambda_c$ and $\sum_{c\in\mathcal{C}_{r(j)}} u_j^c \lambda_c$ are determined by the selected sequences. Constraint (4) ensures that exactly one of the two directions is active. These two constraints couple departures of the same ramp across different runways and therefore cannot be encoded within any single-runway sequence. The pricing subproblem for runway r seeks the sequence with the negative reduced cost. The reduced cost involves the dual multiplier of the covering constraint (2) and the dual multipliers of the pushback interval constraints (3), which attach a cost to the pushback time of each departure contained in the column. Each runway yields only one selected column, and the dual signal available to the pricing subproblems is therefore weak.

2 Details of the Lagrangian Relaxation Formulation

For the RSPCP, the only constraints that couple the different runways are the ramp-level pushback interval constraints.

$$U_j - U_i \geq \zeta_p + \Pi_{ij}(1 - K_{ij}^p), \quad \forall p \in \mathcal{P}, \forall i,j \in \mathcal{D}_p : i \neq j.$$

Relaxing these constraints with multipliers $\mu_{ij}^p \geq 0$ yields the Lagrangian function

$$\begin{aligned}\mathcal{L}(V, H, U, Z, K; \mu) = &\sum_{i \in \mathcal{D}} c_i^d (V_i - e_i^d) + \sum_{i \in \mathcal{A}} c_i^a (H_i - e_i^a) \\ &+ \sum_{p \in \mathcal{P}} \sum_{\substack{i,j \in \mathcal{D}_p \\ i \neq j}} \mu_{ij}^p \big[\zeta_p + \Pi_{ij}(1 - K_{ij}^p) - U_j + U_i\big],\end{aligned} \tag{7}$$

and the Lagrangian dual is $\max_{\mu \geq 0} g(\mu)$ with $g(\mu) = \min \mathcal{L}(\cdot; \mu)$ over the remaining constraints. For a fixed μ, the K variables separate from the remaining variables. For each unordered pair $\{i, j\}$, the terms involving K reduce to

$$\begin{aligned}\mu_{ij}^p \Pi_{ij}(1 - K_{ij}^p) + \mu_{ji}^p \Pi_{ji}(1 - K_{ji}^p) &= \mu_{ij}^p \Pi_{ij} + \mu_{ji}^p \Pi_{ji} - \mu_{ij}^p \Pi_{ij} K_{ij}^p - \mu_{ji}^p \Pi_{ji} K_{ji}^p \\ &= \mu_{ij}^p \Pi_{ij} + K_{ij}^p \big(\mu_{ji}^p \Pi_{ji} - \mu_{ij}^p \Pi_{ij}\big),\end{aligned}$$

where the second equality uses $K_{ji}^p = 1 - K_{ij}^p$. Since $K_{ij}^p \in \{0, 1\}$, minimizing over the two possible values gives

$$\min_{K_{ij}^p \in \{0,1\}} \Big[\mu_{ij}^p \Pi_{ij} + K_{ij}^p \big(\mu_{ji}^p \Pi_{ji} - \mu_{ij}^p \Pi_{ij}\big)\Big] = \min\big\{\mu_{ij}^p \Pi_{ij},\, \mu_{ji}^p \Pi_{ji}\big\},$$

which yields the closed form $K_{ij}^p = 1$ whenever $\mu_{ij}^p \Pi_{ij} \geq \mu_{ji}^p \Pi_{ji}$ and $K_{ji}^p = 1$ otherwise. The relaxed problem accordingly decomposes by runway, and the dual function splits as

$$g(\mu) = \sum_{r \in \mathcal{R}} g_r(\mu) + \sum_{p \in \mathcal{P}} \sum_{\substack{i,j \in \mathcal{D}_p \\ i \neq j}} \mu_{ij}^p \zeta_p + \sum_{p \in \mathcal{P}} \sum_{\substack{i<j \\ i,j \in \mathcal{D}_p}} \min\{\mu_{ij}^p \Pi_{ij},\, \mu_{ji}^p \Pi_{ji}\}, \tag{8}$$

where the last sum is the result of the minimization above, summed over all unordered pairs $\{i, j\}$ in $\mathcal{D}_p$. In (8), $g_r(\mu)$ is the subproblem of runway r, which minimizes the runway-level objective

$$\sum_{i \in \mathcal{D}^r} c_i^d (V_i - e_i^d) + \sum_{i \in \mathcal{A}^r} c_i^a (H_i - e_i^a) + \sum_{i \in \mathcal{D}^r} \Big(\sum_{\substack{j \in \mathcal{D}_{p(i)} \\ j \neq i}} \mu_{ij}^{p(i)} - \sum_{\substack{j \in \mathcal{D}_{p(i)} \\ j \neq i}} \mu_{ji}^{p(i)} \Big) U_i \tag{9}$$

subject to the runway sequencing constraints, the time window constraints, and the linking constraints $V_i = U_i + \delta_{pr}$, where $p(i)$ denotes the ramp of aircraft i. Substituting $U_i = V_i - \delta_{pr}$ shows that the last term of the objective is a linear cost on the take-off times V_i up to a constant. The subproblem is therefore a single-runway scheduling problem with costs attached to pushback times and can be solved by the same DP as in the DP-BnC. The bound obtained from the dual is generally infeasible for the RSPCP, and recovering a feasible schedule requires a repair operator, such as the sequential repair approach in our DP-BnC. The multipliers are updated by a subgradient scheme, for example $\mu_{ij}^p \leftarrow \max\{0,\, \mu_{ij}^p + \theta_t s_{ij}^p\}$ with the subgradient $s_{ij}^p = \zeta_p + \Pi_{ij}(1 - K_{ij}^p) - U_j + U_i$ and a stepsize θ_t, or by more advanced schemes such as surrogate and incremental Lagrangian relaxation [1].

References

[1] Bragin, M. A. (2024). Survey on Lagrangian relaxation for MILP: Importance, challenges, historical review, recent advancements, and opportunities. *Annals of Operations Research*, 333(1):29–45.

3 Details of the Logic-Based Benders Decomposition

We first illustrate the difference between the no-good cut and the logic-based feasibility cut with the example in Figure 1. Suppose runway 1 is assigned aircraft 1, 2, 3, and 4, and runway 2 is assigned aircraft 5, 6, 7, and 8, and the feasible sequencing solutions of the two runways are depicted in the upper part of Figure 1(a). In some iteration, the master problem returns the sequences 2 (20), 1 (80), 3 (200), 4 (275) and 5 (36), 8 (118), 7 (178), 6 (253), where the numbers in parentheses denote the runway times, and the pushback interval between aircraft 1 and 8 is violated, as highlighted in the lower left part of Figure 1(a). As shown in the lower right part of Figure 1(a), a no-good cut only tells the master problem that this complete runway sequence is infeasible and should be avoided in later iterations. In contrast, as shown in Figure 1(b), the feasibility cut we use, following the idea of the state-based cut, fixes the problematic partial sequences 2 (20), 1 (80) and 5 (36), 8 (118) highlighted in green, and tells the master problem that any solution produced later cannot simultaneously contain these two partial sequences, as illustrated by the multiple panels in Figure 1(b). As the number of aircraft grows, such cuts eliminate a large number of infeasible solutions.

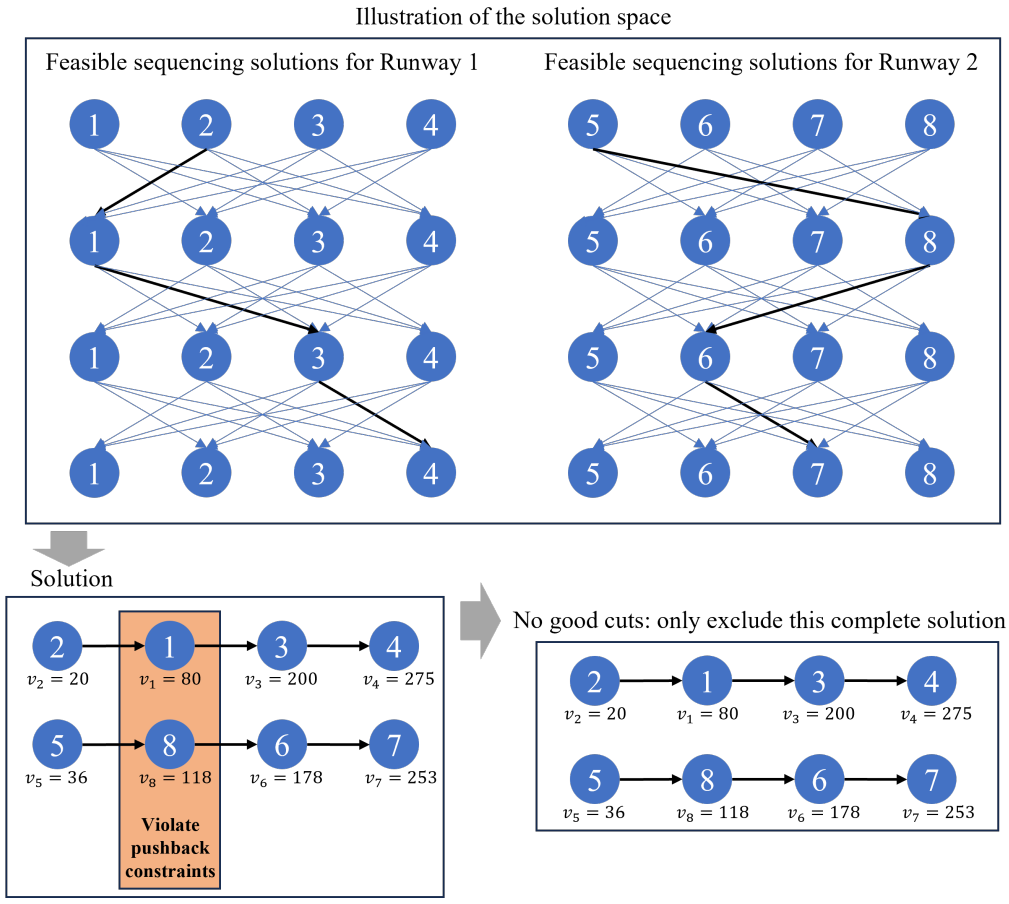
We accordingly design a logic-based Benders decomposition with feasibility cuts to solve the RSPCP. The master problem is formulated as follows.

$$\min_{\boldsymbol{V},\boldsymbol{H},\boldsymbol{U},\boldsymbol{Z},\boldsymbol{K}} \sum_{i\in\mathcal{D}} c_i^d \left(V_i - e_i^d\right) + \sum_{i\in\mathcal{A}} c_i^a \left(H_i - e_i^a\right) \tag{10}$$

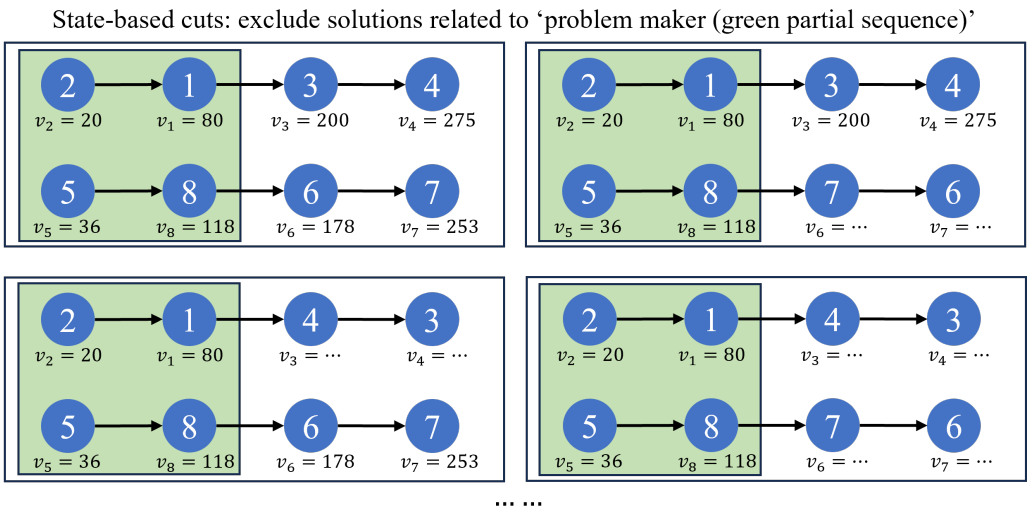
subject to

$$V_i \geq \tau_r + \alpha_{m_r^0 m_i}, \qquad \forall r \in \mathcal{R}, \forall i \in \mathcal{D}^r \tag{11}$$
$$H_i \geq \tau_r + \alpha_{m_r^0 m_i}, \qquad \forall r \in \mathcal{R}, \forall i \in \mathcal{A}^r \tag{12}$$
$$Z_{ij}^r + Z_{ji}^r = 1, \qquad \forall r \in \mathcal{R}, \forall i,j \in \mathcal{N}^r : i \neq j \tag{13}$$
$$Z_{ij}^r = 1, \qquad \forall r \in \mathcal{R}, \forall i,j \in \mathcal{D}^r : i \neq j, m_i = m_j, e_i^d < e_j^d \tag{14}$$
$$Z_{ij}^r = 1, \qquad \forall r \in \mathcal{R}, \forall i,j \in \mathcal{A}^r : i \neq j, m_i = m_j, e_i^a < e_j^a \tag{15}$$
$$V_i + \alpha_{m_i m_j} \leq V_j + \Gamma_{ij}\left(1 - Z_{ij}^r\right), \qquad \forall r \in \mathcal{R}, \forall i,j \in \mathcal{D}^r : i \neq j \tag{16}$$
$$H_i + \alpha_{m_i m_j} \leq H_j + \Gamma_{ij}\left(1 - Z_{ij}^r\right), \qquad \forall r \in \mathcal{R}, \forall i,j \in \mathcal{A}^r : i \neq j \tag{17}$$
$$H_i + \alpha_{m_i m_j} \leq V_j + \Gamma_{ij}\left(1 - Z_{ij}^r\right), \qquad \forall r \in \mathcal{R}, \forall i \in \mathcal{A}^r, \forall j \in \mathcal{D}^r \tag{18}$$
$$V_i + \alpha_{m_i m_j} \leq H_j + \Gamma_{ij}\left(1 - Z_{ij}^r\right), \qquad \forall r \in \mathcal{R}, \forall i \in \mathcal{D}^r, \forall j \in \mathcal{A}^r \tag{19}$$
$$V_i = U_i + \delta_{pr}, \qquad \forall r \in \mathcal{R}, \forall p \in \mathcal{P}, \forall i \in \mathcal{D}_p^r \tag{20}$$
$$K_{ij}^p + K_{ji}^p = 1, \qquad \forall p \in \mathcal{P}, \forall i,j \in \mathcal{D}_p : i \neq j \tag{21}$$
$$e_i^d \leq V_i \leq l_i^d, V_i \in \mathbb{R}_+, \quad U_i \in \mathbb{R}_+, \qquad \forall i \in \mathcal{D} \tag{22}$$
$$e_i^a \leq H_i \leq l_i^a, H_i \in \mathbb{R}_+, \qquad \forall i \in \mathcal{A} \tag{23}$$
$$Z_{ij}^r \in \{0,1\}, \qquad \forall r \in \mathcal{R}, \forall i,j \in \mathcal{N}^r : i \neq j \tag{24}$$
$$K_{ij}^p \in \{0,1\}. \qquad \forall p \in \mathcal{P}, \forall i,j \in \mathcal{D}_p : i \neq j. \tag{25}$$

The master problem retains the runway scheduling decisions and the pushback-related variables U and K, together with constraints (20) and (21), while the pushback interval constraints are removed, so that the master problem is a relaxation of the original model. These variables and constraints do not increase the difficulty of the master problem. Since U and K do not enter the objective or the runway sequencing constraints, constraints (20) and (21) are inactive with respect to the optimal value of the master problem. They are retained as a carrier for the feasibility cuts, so that the cut information can be expressed and communicated through these variables. We solve the master problem using GUROBI to obtain a tentative solution, including the runway sequence and the take-off and landing times on each runway. For the subproblem,



(a) Solution space and the no-good cut for a violated pushback interval



(b) State-based cut excludes all solutions containing the green partial sequences

Figure 1: Illustration of the difference between the no-good cut and the state-based cut.

we apply the same feasibility check used in the DP-BnC framework and, whenever the solution is infeasible, identifies the partial sequences on the involved runways that jointly violate a pushback interval. Suppose the violation is traced to two partial sequences on runways r_a and r_b. Let $S_{r_a} = \{a_1, a_2, \ldots, a_k\}$ be the partial sequence on runway r_a and $S_{r_b} = \{b_1, b_2, \ldots, b_l\}$ that on runway r_b, each ordered by its take-off or landing times, where a_k and b_l belong to the same ramp and their pushback times are too close. We add the following feasibility cut, which requires that whenever the two partial sequences appear simultaneously, the pushback interval between a_k and b_l holds in one of the two directions

$$U_j - U_i \geq \zeta_p + \Pi_{ij}\left(1 - K^p_{ij}\right) - M\,\Phi(S), \qquad \forall i, j \in \{a_k, b_l\}, i \neq j, \tag{26}$$

where S denotes the combination of the two partial sequences and

$$\Phi(S) = \sum_{h=1}^{k-1}\left(1 - Z^{r_a}_{a_h a_{h+1}}\right) + \sum_{h=1}^{l-1}\left(1 - Z^{r_b}_{b_h b_{h+1}}\right). \tag{27}$$

The index set $\{i, j\} = \{a_k, b_l\}$ with $i \neq j$ yields the two directions, and since $K^p_{ij} + K^p_{ji} = 1$, exactly one of the two inequalities is active when $\Phi(S) = 0$, depending on the pushback order, so that both possibilities of which aircraft pushes back first are covered. In a multi-runway system, the feasibility check may detect several conflicting pairs of partial sequences at a time. Following the same logic as in the DP-BnC framework, we add only one cut per iteration, corresponding to the earliest conflicting pair. To identify this pair, we compare the aircraft involved in each detected conflict and select the pair whose first violating aircraft has the earliest operation time. The exact choice among the detected pairs is not critical, as long as an early conflict is selected, since the subproblem is re-run and the remaining conflicts are handled in subsequent iterations. Each cut therefore excludes the corresponding combination of two partial sequences, and the procedure iterates until no violation is detected.